



JANUARY 18, 2025

WEB OF DATA

PROJECT

DHEERAJ PARKASH

M2 Data Science



Introduction

The motivation behind this exploration is to understand the interaction between cultural heritage locations and their surrounding urban environments. By analyzing the proximity of cities to these sites and demographic factors, we can gain insights into accessibility, potential human impact, and regional development opportunities. This approach aligns with broader goals of promoting heritage preservation, enhancing cultural tourism, and supporting informed planning for infrastructure and resource allocation.

Dataset

The dataset extracted from [UNESCO World Heritage List](#) contains comprehensive information about heritage sites, including unique identifiers, names in multiple languages, descriptions, justifications for inclusion, inscription dates, geographical coordinates (longitude, latitude), area in hectares, and associated criteria and categories.

From this, a Python script was used to create a simplified CSV file containing only the relevant fields: `unique_number` (site identifier), `iso_code` (country code), `name_en` (English site name), `longitude`, `latitude`, `states_name_en` (country names), `region_en` (regional classification), and `coordinates` (latitude-longitude pair). This reduced dataset focuses on essential attributes for spatial analysis and integration into applications like proximity calculations and regional insights.

Convert to a Knowledge Graph

Using these prefixes

PREFIX un: <<https://paris-saclay.fr/project/un/>> .

PREFIX unhs: <<https://paris-saclay.fr/project/un/historical-site/>> .

PREFIX loc: <<https://paris-saclay.fr/project/un/historical-site/location/>> .

PREFIX geo: <<http://www.opengis.net/ont/geosparql#>> .

PREFIX owl: <<http://www.w3.org/2002/07/owl#>> .

PREFIX rdfs: <<http://www.w3.org/2000/01/rdf-schema#>> .

Mappings

Name: unhsite csvIdentifier: 2212558857085

ConfigurationPreviewBoth

All mapping changes savedSaveDownload JSONUpload JSONRDFOpen in GraphDBNew Mapping

unique_numberiso_codename_enlongitudelatitudestates_name_enregion_encoordinates

unique_number

geo ×loc ×owl ×rdfs ×un ×unhs ×

unhs: unique_number<IRI>	a	un: HistoricalSite<IRI>
		geo: Feature<IRI>
	rdfs: label<IRI>	name_en"Literal"@Languageen
	geo: hasGeometry<IRI>	loc: unique_number<IRI>
loc: unique_number<IRI>	geo: asWKT<IRI>	coordinates"Literal"^^Datatype

Queries

1. On UN historical site RDF data to get location and name of site

```
PREFIX geo: <http://www.opengis.net/ont/geosparql#>
PREFIX un: <https://paris-saclay.fr/project/un/>
PREFIX unhs: <https://paris-saclay.fr/project/un/historical-site/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX loc: <https://paris-saclay.fr/project/un/historical-site/location/>

SELECT ?s ?cord ?name ?loc
WHERE {
  ?s a un:HistoricalSite;
    rdfs:label ?name;
    geo:hasGeometry ?cord .
  ?cord geo:asWKT ?loc
}
```

2. Data of French Cities from wikidata

```
PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX wd: <http://www.wikidata.org/entity/>
PREFIX wdt: <http://www.wikidata.org/prop/direct/>
PREFIX wikibase: <http://wikiba.se/ontology#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX geo: <http://www.opengis.net/ont/geosparql#>
PREFIX shp: <https://paris-saclay.fr/courses/wod/shape/>
PREFIX wod: <https://paris-saclay.fr/courses/wod/>

SELECT ?citywikidata ?wikidataCityLabel ?population ?coordinates
WHERE {
  # Federated query to Wikidata
  SERVICE <https://query.wikidata.org/sparql> {
    ?citywikidata wdt:P31 wd:Q515; # Instance of city
      wdt:P17 wd:Q142; # Located in France
      rdfs:label ?wikidataCityLabel;
      wdt:P1082 ?population; # Population claim
      wdt:P625 ?coordinates. # Location coordinates (latitude, longitude)

    FILTER(LANG(?wikidataCityLabel) = "en") # Match French city labels
  }
}
```

3. Get the data of French cities with the 100km of historical sites

```
PREFIX geo: <http://www.opengis.net/ont/geosparql#>
PREFIX un: <https://paris-saclay.fr/project/un/>
PREFIX unhs: <https://paris-saclay.fr/project/un/historical-site/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX loc: <https://paris-saclay.fr/project/un/historical-site/location/>
PREFIX wd: <http://www.wikidata.org/entity/>
PREFIX wdt: <http://www.wikidata.org/prop/direct/>
PREFIX wikibase: <http://wikiba.se/ontology#>
PREFIX shp: <https://paris-saclay.fr/courses/wod/shape/>
PREFIX wod: <https://paris-saclay.fr/courses/wod/>
PREFIX uom: <http://www.opengis.net/def/uom/OGC/1.0/>
PREFIX geof: <http://www.opengis.net/def/function/geosparql/>

SELECT ?citywikidata ?wikidataCityLabel ?population ?coordinates ?heritageSite ?siteLabel
?siteCoordinates
WHERE {
  # Local query for historical sites and their coordinates
  ?heritageSite a un:HistoricalSite;
    rdfs:label ?siteLabel;
    geo:hasGeometry ?siteCord.
  ?siteCord geo:asWKT ?siteCoordinates.

  # Federated query to Wikidata for cities within France
  SERVICE <https://query.wikidata.org/sparql> {
    ?citywikidata wdt:P31 wd:Q515; # Instance of city
      wdt:P17 wd:Q142; # Located in France
      rdfs:label ?wikidataCityLabel;
      wdt:P1082 ?population; # Population claim
      wdt:P625 ?coordinates. # Location coordinates (latitude, longitude)

    FILTER(LANG(?wikidataCityLabel) = "en") # Match French city labels
  }

  # Calculate the distance between heritage site and city
  BIND(geof:distance(?siteCoordinates, ?coordinates, uom:metre) AS ?distance)

  # Filter cities within 100 km (100000 meters)
  FILTER(?distance <= 100000)

}
#LIMIT 100
ORDER BY ?siteLabel ?population
```

4. Get the total of population of cities around 100km of UN historical site

```
PREFIX geo: <http://www.opengis.net/ont/geosparql#>
PREFIX un: <https://paris-saclay.fr/project/un/>
PREFIX unhs: <https://paris-saclay.fr/project/un/historical-site/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX loc: <https://paris-saclay.fr/project/un/historical-site/location/>
PREFIX wd: <http://www.wikidata.org/entity/>
PREFIX wdt: <http://www.wikidata.org/prop/direct/>
PREFIX wikibase: <http://wikiba.se/ontology#>
PREFIX shp: <https://paris-saclay.fr/courses/wod/shape/>
PREFIX wod: <https://paris-saclay.fr/courses/wod/>
PREFIX uom: <http://www.opengis.net/def/uom/OGC/1.0/>
PREFIX geof: <http://www.opengis.net/def/function/geosparql/>

SELECT ?heritageSite ?siteLabel ?siteCoordinates (SUM(?population) AS ?totalPopulation)
WHERE {
  # Local query for historical sites and their coordinates
  ?heritageSite a un:HistoricalSite;
    rdfs:label ?siteLabel;
    geo:hasGeometry ?siteCord.
  ?siteCord geo:asWKT ?siteCoordinates.

  # Federated query to Wikidata for cities within France
  SERVICE <https://query.wikidata.org/sparql> {
    ?citywikidata wdt:P31 wd:Q515; # Instance of city
      wdt:P17 wd:Q142; # Located in France
      rdfs:label ?wikidataCityLabel;
      wdt:P1082 ?population; # Population claim
      wdt:P625 ?coordinates. # Location coordinates (latitude, longitude)

    FILTER(LANG(?wikidataCityLabel) = "en") # Match French city labels
  }

  # Calculate the distance between heritage site and city
  BIND(geof:distance(?siteCoordinates, ?coordinates, uom:metre) AS ?distance)

  # Filter cities within 100 km (100000 meters)
  FILTER(?distance <= 150000)
}
GROUP BY ?heritageSite ?siteLabel ?siteCoordinates
ORDER BY ?totalPopulation
```

5. To visualize the data for location: Point(2.301666667 49.895) place name: Amiens Cathedral, on <https://yasgui.triply.cc/>

```
PREFIX geo: <http://www.opengis.net/ont/geosparql#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX wd: <http://www.wikidata.org/entity/>
PREFIX wdt: <http://www.wikidata.org/prop/direct/>

SELECT ?citywikidata ?wikidataCityLabel ?population ?coordinates
WHERE {
  SERVICE <https://query.wikidata.org/sparql> {
    ?citywikidata wdt:P31 wd:Q515; # Instance of city
    wdt:P17 wd:Q142; # Located in France
    rdfs:label ?wikidataCityLabel;
    wdt:P1082 ?population; # Population
    wdt:P625 ?coordinates. # Coordinates

    FILTER(LANG(?wikidataCityLabel) = "en") # English labels

    # Extract latitude and longitude from coordinates
    BIND(STRATELTER(STR(?coordinates), "Point(") AS ?coordString)
    BIND(STRBEFORE(?coordString, "(") AS ?coords)
    BIND(xsd:float(STRBEFORE(?coords, " ")) AS ?lon)
    BIND(xsd:float(STRATELTER(?coords, " ")) AS ?lat)

    # Center point (Amiens Cathedral): 49.895, 2.301666667
    # Bounding box for 150 km (~1.35 degrees latitude, ~2.25 degrees longitude at this latitude)
    FILTER(?lat > 49.895 - 1.35 && ?lat < 49.895 + 1.35) # Latitude range
    FILTER(?lon > 2.301666667 - 2.25 && ?lon < 2.301666667 + 2.25) # Longitude range
  }
}
ORDER BY DESC(?population)
```



